LayoutAD: Exploring Semantic-Geometric Misalignment Reasoning for Scene Layout Anomaly Detection

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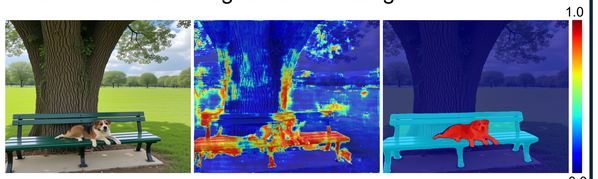
CVPR JUNE 3-7, 2026 DENVER COLORADO

Motivation

Existing anomaly detection methods mainly focus on pixel-level defects, but generative hallucinations in natural scenes often appear as high-level semantic inconsistencies and geometric misalignments.



Input Image Pixel-level AD Object-level AD

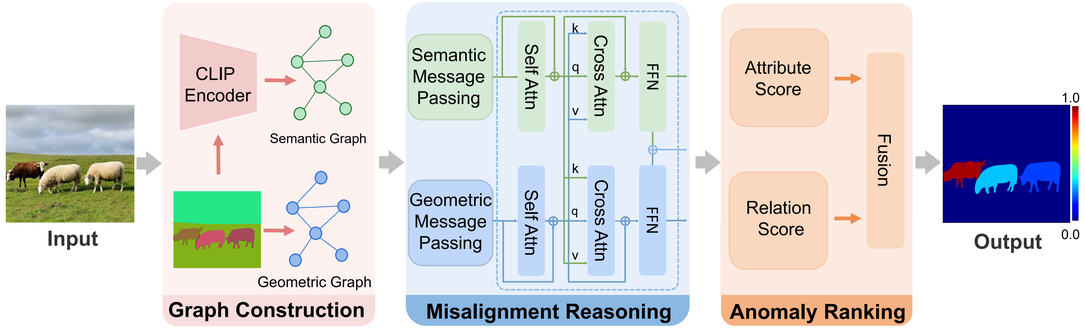
Layout Anomaly Detection

- Given a natural scene image, ours goal is to detect anomalous objects or relations with semantic and geometric misalignments.
- The detected scene layout anomalies can support various downstream scene understanding and image generation.

Contributions

- We make the first attempt to investigate the new problem of scene layout anomaly detection.
- We propose an unsupervised framework by explicitly modeling semantic and geometric interactions among objects in the scene.
- Experimental results show that our model effectively detect diverse layout anomalies and enables downstream applications.

Approach



Graph Construction Module

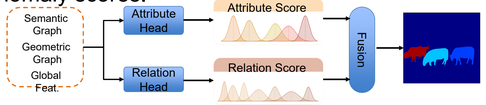
GCM builds complementary semantic and geometric graphs to represent object appearances and spatial layouts.

Misalignment Reasoning Module

MRM performs message passing for local graph structures and cross-graph attention to detect layout inconsistencies.

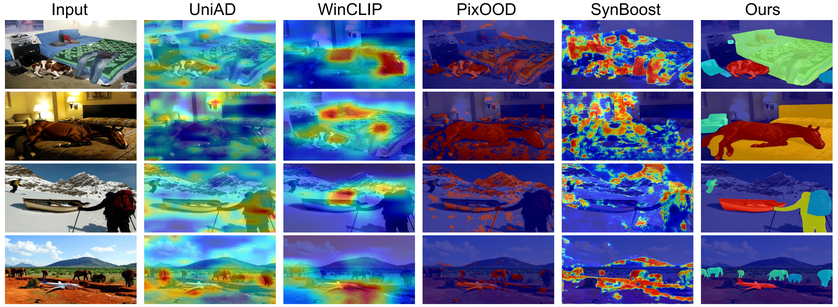
Anomaly Ranking Module

ARM uses density estimators to compute and fuse attribute and relation anomaly scores.



Experiments

Qualitative & Quantitative Evaluation



Metric	PatchCore	SimpleNet	UniAD	GeneralAD	SynBoost	PixOOD	Ours
I-AUROC ↑	0.539	0.551	0.479	0.543	0.542	0.538	**0.586**
P-AUROC↑	0.571	0.571	0.575	0.565	0.773	0.720	**0.871**
AP-AUROC↑	0.565	0.515	0.508	0.314	0.777	0.722	**0.883**

Application

Self-corrected Image Generation

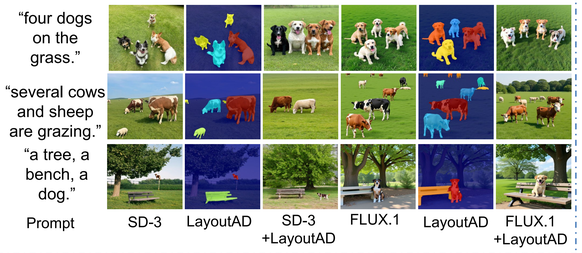
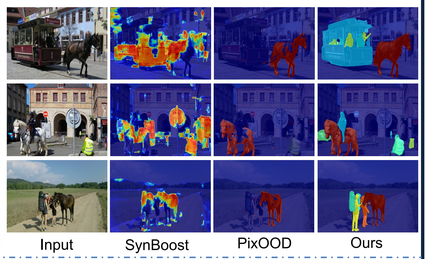


Image Anomaly Segmentation



Video Anomaly Detection